ASSIGNMENT SUBMISSION • TASK 3

Metric Definition and Data Quality Report

Unambiguous definitions, edge case documentation, and data quality audit report.

1. Unambiguous Metric Definitions

Metric A: Monthly Active Users (MAU)

Definition: Total count of unique authenticated users who perform at least one intentional, qualifying user action within a specific calendar month.

```
MAU = Count(Distinct User_ID) where Event_Timestamp in Month AND Is_Qualifying = True
```

- **Edge Cases:** Anonymous/unauthenticated visitors are excluded. Passive background pings or email opens do not count as activity. All timestamps normalized to UTC.

Metric B: Net Recurring Revenue (NRR)

Definition: Percentage of recurring revenue retained from existing customers over a period, including expansions, contractions, and churn, excluding new acquisitions.

```
NRR = [(Starting ARR + Expansion ARR - Contraction ARR - Churn ARR) / Starting ARR] × 100
```

- **Edge Cases:** New customers acquired during the month are strictly excluded. One-time setup credits do not alter ARR. Pending cancellations apply when the contract legally expires.

Metric C: Average Order Value (AOV)

Definition: Average gross revenue generated per completed, non-canceled order placed on the platform during a specified timeframe.

```
AOV = ∑ (Gross Order Value - Item Discounts) / Total Completed Orders
```

- **Edge Cases:** Fully refunded/canceled orders before fulfillment are removed. Taxes and customer-paid shipping are excluded. Zero-dollar promotional orders ($0) are excluded.

2. Data Quality Audit Report

Audited Dataset: `fact_orders` & `dim_users` (Sample Size: 500,000 records)

Quality Issue	Field Affected	Identified Anomaly	Error Count	Impacted %
Null Values	`user_id`	Missing user identifier on completed orders	4,120	0.82%
Duplicates	`order_id`	Duplicate transaction logs via webhook retries	1,850	0.37%
Impossible Values	`order_total`	Negative value (< $0) or test value (> $50,000)	312	0.06%
Timezone Mismatch	`created_at`	Unconverted local time strings mixed with UTC	12,400	2.48%
Orphaned Records	`payment_status`	Marked 'Completed' without payment log	540	0.11%

3. Skew Analysis: Impact of Quality Issues on Metrics

1. **Null `user_id` on MAU:** Aggregating null values under a single ID depresses active user count, while treating each null as unique artificially inflates MAU.

2. **Duplicate `order_id` on AOV:** Duplicate webhook rows double-count sales revenue and completed order volume, distorting both numerator and denominator in AOV calculation.

3. **Impossible/Outlier `order_total` on AOV:** Negative totals reduce revenue falsely, while extreme test orders (e.g., $999,999) severely skew the mean AOV upwards.

4. **Timezone Mismatches on Monthly Metrics:** Unconverted local timestamps push end-of-month transactions into the wrong calendar month, creating artificial revenue swings in NRR and MAU reporting.